Notes: Grinblatt, Keloharju, and Linnainmaa (JF, 2011)

IQ and Stock Market Participation

Contents

1 Big picture

- **Question.** Does cognitive ability help explain why many households stay out of the stock market, even when standard finance theory predicts nearly universal participation?
- **Core puzzle.** Fixed participation costs can explain some nonparticipation among poor households, but they struggle to explain why many affluent households also avoid stocks. The paper asks whether low cognitive skill acts like an important information-processing friction.
- **Main idea.** IQ is treated as a proxy for the ability to process financial information, understand the risk-return trade-off, and avoid investment mistakes. If cognitive skill matters, stock market participation should rise with IQ even after controlling for wealth, income, education, and demographics.
- **Data advantage.** The authors combine Finnish military IQ records, exact administrative records of stock ownership, mutual fund holdings from tax files, detailed tax-return controls, and historical address data that identify siblings. This avoids the usual survey-response and self-reporting problems.
- **Main participation result.** Participation is strongly and monotonically increasing in IQ. In the benchmark probit, the participation rate for the lowest IQ stanine is **20.5 percentage points** lower than for the highest IQ stanine, conditional on a rich set of controls.
- **Why this is striking.** The IQ effect is economically large - larger than the effect of income in the same regressions - and it remains strong even among the top decile of the income or wealth distribution.
- **Secondary-channel result.** Part of the IQ gradient works through variables that IQ itself helps determine, especially wealth, education, and income. But even after accounting for these channels, a substantial direct IQ effect remains.
- **Family and endogeneity result.** Sibling-based evidence suggests the relation is not just family background or omitted-variable bias. A control-function approach that instruments own IQ with a brother's IQ and a family random-effects probit both continue to find a positive own-IQ effect.
- **Conditional-on-participation result.** High-IQ participants hold more mutual funds, own more stocks, bear less idiosyncratic and systematic risk, and earn higher Sharpe ratios. The paper therefore links IQ not only to market entry but also to portfolio quality.
- **Economic interpretation.** Low-IQ households may rationally stay out because they face a worse effective investment opportunity set: poorer diversification, more mistakes, and a worse risk-return trade-off.
- **Takeaway.** The participation puzzle is not just about entry fees or taste heterogeneity. Cognitive skill appears to be a first-order determinant of both whether households enter the stock market and how well they invest once they do.

2 Data and measurement (key objects)

2.1 Six merged data sets

The paper merges six administrative data sources:

- **Finnish Central Securities Depository (FCSD).** Exact daily holdings and trades of all Finnish household investors in FCSD-registered stocks from 1995 to 2002.
- **Helsinki Exchanges (HEX).** Daily prices and returns for stocks traded on the Helsinki Exchanges, used to construct portfolio risk measures.
- **Finnish Tax Administration (FTA).** Year-2000 tax returns and registry information with income, taxable net worth, homeownership, foreign assets, private business ownership, marital status, children, employment status, occupation, gender, and language.
- **Finnish Armed Forces (FAF).** IQ scores from the intelligence battery taken around entry into mandatory military service.
- **Finnish address records.** Historical residential locations, which let the authors identify sibling pairs.
- **Finnish census data.** Zip-code-by-age-group education shares, used as a proxy for individual education.

2.2 Main sample

The core sample contains **158,044 males** who:

- took the military IQ exam between 1982 and 2001,
- were born between 1953 and 1982,
- lived in Uusimaa or East Uusimaa in year 2000, and
- have complete tax and education-proxy information.

The paper also studies:

- **4,358 sisters** identified from the address data, and
- **1,996 brother pairs** for the sibling-IV and family-effects exercises.

Because the sample comes from the Greater Helsinki area, it is slightly more affluent and slightly higher IQ than the national male population, but still very large and heterogeneous.

2.3 IQ measurement

The FAF intelligence battery contains **120 questions** that assess:

- mathematical ability,
- verbal ability,
- logical ability.

The armed forces aggregate these into a composite intelligence score, which the paper calls **IQ**. IQ is reported on a **stanine scale** from 1 to 9, where 9 is the highest cognitive-skill category.

A useful interpretation is that adjacent stanines differ by about half a standard deviation around the center of the distribution. The low and high tails are extreme: stanine 1 and stanine 9 correspond to subjects well into the left and right tails of ability.

2.4 Participation variable

The main outcome is a stock market participation dummy:

$$D_i = \mathbf{1}\{\text{household } i \text{ holds FCSD stocks or mutual funds on 31 Dec 2000}\}. \tag{1}$$

This definition includes direct stock ownership and indirect participation through mutual funds. The authors also run robustness checks with narrower definitions:

- holding at least one individual stock,
- holding at least two different stocks,
- holding a mutual fund but no stock,
- purchasing stock with cash rather than receiving it through compensation-related channels.

2.5 Controls

The regressions use a rich set of controls built from the tax and census data:

- income deciles and income growth,
- taxable net-worth deciles,
- dummies for housing, forest ownership, private equity ownership, and foreign assets,
- language, marital status, cohabitation, and children,
- occupation dummies including entrepreneur, farmer, and finance professional,
- unemployment status,
- cohort fixed effects,
- zip-code-by-age-group education shares.

The baseline specification contains **67 control regressors**, and inference uses zip-code clustered standard errors.

2.6 Portfolio-quality outcomes

For participants, the paper studies several outcome variables related to risk and performance:

- estimated Sharpe ratio of the risky portfolio,

- total portfolio volatility,
- stock-portfolio variance,
- residual variance from a market model,
- number of stocks held,
- mutual fund ownership,
- systematic variance,
- beta, size rank, and book-to-market rank of the stock portfolio.

Because the data do not identify which mutual funds each person holds, the paper imputes the mutual-fund component using the HEX index portfolio.

2.7 Key descriptive patterns

The raw data already show a strong IQ gradient:

- participants have an average IQ almost **one full stanine** above nonparticipants,
- participants earn about **29,604 euros** of labor income on average versus **19,901 euros** for nonparticipants,
- participation rises almost monotonically from **10%** in stanine 1 to **47%** in stanine 9,
- average income rises from **16,062 euros** in stanine 1 to **31,707 euros** in stanine 9,
- taxable net worth rises from **3,627 euros** to **43,619 euros**,
- homeownership rises from **28%** to **43%**,
- marriage rises from **22%** to **33%**.

These patterns are exactly why the authors move to multivariate regressions: IQ is clearly correlated with many things that also affect participation.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline participation probit

The main specification is a binary-response model:

$$\text{Prob}(D_i = 1 \mid IQ_i, X_i) = \Phi\big(\alpha + \beta IQ_i + \gamma' X_i\big)\,, \tag{2}$$

where:

- D_i is stock market participation,
- IQ_i is either the stanine score or a set of IQ-stanine dummies,

- X_i is the vector of controls,
- $\Phi(\cdot)$ is the standard normal cdf.

The paper estimates two versions:

$$\mathrm{Prob}(D_i = 1 \mid X_i) = \Phi\left(\alpha + \sum_{s=1}^{8} \beta_s \mathbf{1}\{IQ_i = s\} + \gamma' X_i\right), \tag{3}$$

$$\mathrm{Prob}(D_i = 1 \mid X_i) = \Phi(\alpha + \beta IQ_i + \gamma' X_i)\,. \tag{4}$$

Interpretation. The dummy specification is the flexible, nonparametric version that checks whether the IQ effect is monotone across all stanines. The linear specification compresses IQ into one slope coefficient and is convenient for comparing magnitudes across samples and later robustness exercises.

3.2 Affluent-subsample probits

To test whether IQ still matters when fixed participation costs should be least important, the paper re-estimates the same probit on affluent households:

$$\mathrm{Prob}(D_i = 1 \mid IQ_i, X_i, A_i = 1) = \Phi(\alpha + \beta IQ_i + \gamma' X_i)\,, \tag{5}$$

where $A_i = 1$ indicates either:

- income in the top decile, or
- taxable net worth in the top decile.

Interpretation. If IQ mattered only because low-IQ households are poor and cannot cover entry costs, the coefficient should collapse in these wealthy subsamples. It does not.

3.3 Fairlie-Blinder-Oaxaca decomposition

The paper decomposes participation gaps between high- and low-IQ groups into parts explained by differences in observables and an unexplained residual. For two groups H and L, predicted participation rates satisfy

$$\Delta = \bar{P}_H - \bar{P}_L = \left[\frac{1}{N_H}\sum_{i=1}^{N_H} \Phi(X_i^H \hat{\theta})\right] - \left[\frac{1}{N_L}\sum_{i=1}^{N_L} \Phi(X_i^L \hat{\theta})\right]. \tag{6}$$

The authors then replace one covariate block at a time - wealth, education, income, occupation/employment, and so on - to measure each block's contribution to Δ.

Interpretation. This exercise does *not* ask whether IQ matters. It asks *through which observables* IQ matters. Wealth, education, and income turn out to be the main secondary channels.

3.4 Sibling control-function specification

To address omitted-variable bias, the paper uses a brother's IQ as an instrument-like predictor for own IQ. The first stage is

$$IQ_i = \delta_0 + \delta_1 IQ_i^{bro} + \delta_2' X_i + s_i, \tag{7}$$

where IQ_i^{bro} is the brother's IQ and s_i is the first-stage residual.

The second-stage control-function probit is

$$\text{Prob}(D_i = 1 \mid IQ_i, X_i, s_i) = \Phi\big(\alpha + \beta IQ_i + \gamma' X_i + \rho s_i\big) . \tag{8}$$

Interpretation. If unobservables correlated with IQ are driving the baseline results, the residual term should absorb that correlation and the corrected IQ effect should weaken sharply. Instead, the estimated IQ effect remains strong and actually becomes larger.

3.5 Family random-effects probit

The paper also estimates a brother-pair model with a family random effect:

$$\text{Prob}(D_{if} = 1 \mid IQ_{if}, X_{if}, u_f) = \Phi\big(\alpha + \beta IQ_{if} + \gamma' X_{if} + u_f\big) , \tag{9}$$

where u_f is an unobserved family-specific component shared by brothers in family f.

Interpretation. This asks whether IQ still predicts participation *within families*, after common family background is partialled out. The answer is yes.

3.6 Sharpe ratios and risk regressions

For participants, the paper studies the risk-return trade-off using regressions of the form

$$Y_i = \alpha + \beta IQ_i + \gamma' X_i + \varepsilon_i, \tag{10}$$

where Y_i is alternatively the Sharpe ratio, volatility, residual variance, systematic variance, beta, size rank, or book-to-market rank.

The estimated Sharpe ratio is

$$\widehat{SR}_i = \frac{\widehat{\mu}_i - r_f}{\widehat{\sigma}_i}, \tag{11}$$

with the risky portfolio built from observed stocks and an imputed mutual-fund component.

Portfolio variance is decomposed using a one-factor market model:

$$r_{it} - r_{ft} = \alpha_i + \beta_i (r_{Mt} - r_{ft}) + \varepsilon_{it}. \tag{12}$$

Thus,

$$\text{systematic variance}_i = \beta_i^2 \text{Var}(r_M - r_f), \tag{13}$$

$$\text{residual variance}_i = \text{Var}(\varepsilon_i). \tag{14}$$

The number of stocks held is modeled with a negative-binomial regression, and mutual-fund ownership with another probit.

Interpretation. These regressions test whether low-IQ individuals stay out because their actual investing skill is worse. The answer is consistent with that story: high-IQ participants are more diversified and face lower risk per unit of expected return.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the baseline design is unusually strong

- **Timing.** IQ is measured around induction into military service, typically at age 19 or 20, long before the 2000 participation outcome is measured. This makes reverse causality highly implausible.
- **Administrative ownership data.** Participation is measured from registry holdings and tax-reported fund ownership, not self-reported survey answers. That removes an important source of bias: low-IQ subjects misunderstanding or misreporting what they own.
- **Very rich controls.** The tax and census data contain unusually detailed controls for wealth, income, occupation, employment, language, marital status, children, housing, and local education composition.
- **Geographic clustering.** Standard errors are clustered at the zip-code level, which helps when nearby households share unobserved traits.

4.2 Threat 1: omitted education or human-capital differences

A natural concern is that IQ is proxying for unobserved education rather than cognition itself. The paper cannot use individual education records, so it uses zip-code-by-age-group education shares as proxies.

This is imperfect, but the authors show that the main result is robust when IQ itself is also aggregated to the zip-code-by-age-group level. The point is not that this solves the education problem mechanically. Rather, it suggests that measurement error in the education proxies is unlikely to be the whole story.

4.3 Threat 2: employer stock compensation

If high-IQ people work at firms that pay them in stock or options, then measured participation could partly reflect compensation rather than an active investment decision.

The paper responds by redefining participation in ways that should be less contaminated by employer awards:

- mutual fund ownership only,
- holding at least two different individual stocks,

- stock purchases made with cash.

The IQ result survives all of these alternative definitions.

4.4 Threat 3: fixed costs or measurement error in wealth and income

The authors show that the IQ gradient persists in the top income decile and the top wealth decile. This matters because affluent subjects are precisely the ones for whom modest entry costs should be easiest to absorb. It also makes it harder to argue that the result is driven by noisy measurement of wealth or income.

4.5 Threat 4: family background and latent preferences

Family background could jointly determine IQ and participation through parental education, financial literacy, household culture, or inherited preferences.

The paper attacks this in three ways:

- sister participation is predicted by brother IQ,
- a control-function procedure instruments own IQ with a brother's IQ,
- a random-effects probit absorbs family-specific unobservables shared by brothers.

All three exercises point in the same direction: family background matters, but it does not wipe out the role of own IQ.

4.6 How convincing is the sibling instrument?

The sibling design is informative because a brother's IQ is highly relevant for own IQ, but it is not magic. Its credibility depends on the idea that once controls are added, the remaining part of brother IQ affects own participation mainly through own IQ and shared family determinants rather than through some separate direct channel.

The paper's strongest evidence is not any one sibling regression in isolation. It is that *all* of the following point the same way:

- baseline administrative-data probits,
- affluent-subsample results,
- sister evidence,
- control-function estimates,
- within-family random-effects probits,
- portfolio-quality regressions.

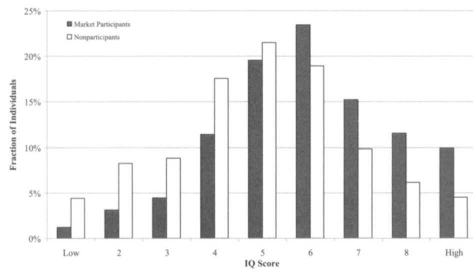

Figure 1. Distribution of IQ score conditional on market participation. Figure 1 plots IQ score distributions for stock market participants and nonparticipants. An individual is a stock market participant if he held mutual funds or individual stocks registered with the FCSD at the end of 2000.

4.7 Important limitations

- The main IQ data are available only for males because they come from military testing.
- Individual education is not observed; only local education composition is.
- Mutual-fund composition is not observed, so Sharpe-ratio calculations require an imputation assumption.
- The sample comes from the Greater Helsinki area and is therefore somewhat more urban and affluent than Finland as a whole.

None of these limitations overturn the main result, but they matter for interpretation and external validity.

5 Tables and figures

5.1 Figure 1: IQ distribution by participation status

Read. Figure 1 plots the distribution of IQ scores separately for participants and nonparticipants.

What it shows. The participant distribution is shifted to the right. There are fewer participants in every below-average IQ stanine and more participants in every above-average IQ stanine.

Economic message. The difference is not driven by a strange spike in one IQ bucket. It is a broad monotone shift in the whole distribution.

5.2 Table I: summary statistics

Table I

Descriptive Statistics

Panel A reports the distribution of IQ scores. Panels B and C report mean values for variables used in regression analyses. See the text for descriptions of the variables. Panel B reports means sorted by participation and Panel C reports means sorted by IQ score. Participation is a dummy variable that takes the value one for subjects who held mutual funds or individual stocks registered with the FCSD at the end of 2000. Income and wealth variables in Panel B are from the 2000 Finnish tax data set. Education variables are derived from the Finnish census data set using each individual's age and zip code. Other demographic and occupation information are from the tax data.

Panel A: Distribution of IQ Score

	IQ Score									
Sample	1	2	3	4	5	6	7	8	9	*N*
Theoretical stanine distribution	4.0%	7.0%	12.0%	17.0%	20.0%	17.0%	12.0%	7.0%	4.0%	
Full IQ score data set	5.2%	9.3%	9.5%	18.4%	21.0%	18.0%	9.1%	5.6%	3.8%	586,187
Uusimaa/East Uusimaa	3.5%	6.8%	7.6%	15.8%	21.0%	20.2%	11.4%	7.7%	6.0%	158,044

Panel B: Mean Socioeconomic Characteristics by Stock Market Participation

		Stock Market Participant	
	All	No	Yes
IQ	5.25	4.97	5.94
Education			
Basic	21.6%	22.4%	19.6%
Vocational	42.6%	43.4%	40.8%
Matricular	18.8%	18.6%	19.5%
University	16.9%	15.6%	20.2%
Ordinary income, EUR	22,642	19,901	29,604
Ordinary income, log-growth	11.8%	11.5%	12.6%
Wealth			
Taxable home wealth >0	37.7%	31.7%	52.8%
Taxable forest wealth >0	1.3%	1.0%	2.0%
Taxable foreign wealth >0	0.0%	0.0%	0.1%

(continued)

Table I—*Continued*

Panel B: Mean Socioeconomic Characteristics by Stock Market Participation

		Stock Market Participant	
	All	No	Yes
Taxable private equity >0	2.6%	2.1%	3.8%
Taxable net worth, EUR	11,193	2,808	32,489
Other demographics			
Swedish	7.0%	6.3%	8.8%
Married	29.6%	27.7%	34.3%
Cohabiter	6.5%	6.9%	5.3%
Kids	29.8%	29.2%	31.5%
Occupation			
Entrepreneur	2.8%	2.9%	2.6%
Farmer	0.9%	0.8%	1.3%
Finance professional	0.7%	0.3%	1.7%
Unemployed	8.6%	10.7%	3.2%
Number of observations	158,044	113,393	44,651

Panel C: Mean Socioeconomic Characteristics by IQ Score

	IQ Score									
	1	2	3	4	5	6	7	8	9	All
Stock market participant	9.8%	13.1%	16.6%	20.4%	26.3%	32.8%	37.9%	42.5%	46.5%	28.3%
Education										
Basic	23.8%	23.6%	23.2%	22.9%	22.0%	21.2%	20.2%	19.5%	18.6%	21.6%
Vocational	47.5%	46.5%	45.9%	44.4%	43.1%	41.7%	40.3%	39.0%	37.2%	42.6%
Matricular	14.1%	15.1%	16.0%	17.3%	18.4%	19.6%	20.9%	22.1%	24.0%	18.8%
University	14.6%	14.8%	14.9%	15.3%	16.5%	17.5%	18.6%	19.5%	20.2%	16.9%
Ordinary income, EUR	16,062	17,666	18,427	19,640	21,413	23,874	26,171	28,191	31,707	22,642
Ordinary income, log-growth	7.1%	7.4%	8.3%	11.0%	11.5%	13.3%	13.4%	14.3%	16.1%	11.8%

(continued)

Table I–*Continued*

Panel C: Mean Socioeconomic Characteristics by IQ Score

	IQ Score									
	1	2	3	4	5	6	7	8	9	All
Wealth										
Taxable home wealth >0	27.9%	31.4%	34.0%	34.8%	37.6%	40.1%	40.8%	42.1%	42.8%	37.7%
Taxable forest wealth >0	1.2%	1.4%	1.2%	1.3%	1.3%	1.2%	1.2%	1.3%	1.4%	1.3%
Taxable foreign wealth >0	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.1%	0.0%
Taxable private equity >0	1.8%	2.1%	2.3%	2.4%	2.4%	2.8%	3.0%	3.0%	3.3%	2.6%
Taxable net worth, EUR	3,627	4,655	7,393	6,730	9,231	9,575	12,019	16,340	43,619	11,193
Other demographics										
Swedish	6.6%	7.4%	10.0%	5.9%	6.7%	6.9%	6.5%	6.9%	8.4%	7.0%
Married	22.5%	25.7%	25.8%	27.0%	29.3%	31.7%	32.0%	34.0%	33.2%	29.6%
Cohabiter	10.1%	10.0%	9.4%	8.0%	6.8%	5.3%	4.3%	3.7%	2.8%	6.5%
Kids	29.7%	32.0%	31.7%	30.5%	30.2%	29.8%	28.3%	28.5%	26.6%	29.8%
Occupation										
Entrepreneur	3.2%	3.6%	3.2%	2.9%	2.7%	2.6%	2.4%	2.5%	2.6%	2.8%
Farmer	1.1%	1.1%	1.2%	0.9%	1.0%	0.8%	0.7%	0.7%	0.7%	0.9%
Finance professional	0.0%	0.1%	0.0%	0.2%	0.6%	0.9%	1.2%	1.7%	1.6%	0.7%
Unemployed	22.4%	16.7%	13.8%	11.3%	8.3%	6.0%	4.2%	3.4%	2.2%	8.6%
Number of observations	5,552	10,749	12,002	25,040	33,124	31,943	17,958	12,145	9,531	158,044

Read. The key summary numbers are:

Statistic	Low IQ (stanine 1)	High IQ (stanine 9)
Participation rate	10%	47%
Average income	16,062 euros	31,707 euros
Taxable net worth	3,627 euros	43,619 euros
Homeownership	28%	43%
Marriage rate	22%	33%

Participants also earn about **29,604 euros** on average versus **19,901 euros** for nonparticipants, and their mean IQ is almost one stanine higher.

Economic message. IQ is strongly correlated with participation, but also with wealth, income, education, employment, and family status. This is why the regression and decomposition exercises matter.

5.3 Table II: baseline probit regressions

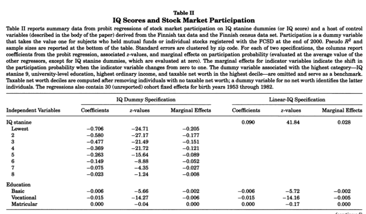

Table II

IQ Scores and Stock Market Participation

Table II reports summary data from probit regressions of stock market participation on IQ stanine dummies (or IQ score) and a host of control variables (described in the body of the paper) derived from the Finnish tax data and the Finnish census data set. Participation is a dummy variable that takes the value one for subjects who held mutual funds or individual stocks registered with the FCSD at the end of 2000. Pseudo R^2 and sample sizes are reported at the bottom of the table. Standard errors are clustered by zip code. For each of two specifications, the columns report coefficients from the probit regression, associated z-values, and marginal effects on participation probability (evaluated at the average value of the other regressors, except for IQ stanine dummies, which are evaluated at zero). The marginal effects for indicator variables indicate the shift in the participation probability when the indicator variable changes from zero to one. The dummy variable associated with the highest category—IQ stanine 9, university-level education, highest ordinary income, and taxable net worth in the highest decile—are omitted and serve as a benchmark. Taxable net worth deciles are computed after removing individuals with no taxable net worth; a dummy variable for no net worth identifies the latter individuals. The regressions also contain 30 (unreported) cohort fixed effects for birth years 1953 through 1982.

	IQ Dummy Specification			Linear-IQ Specification		
Independent Variables	Coefficients	z-values	Marginal Effects	Coefficients	z-values	Marginal Effects
IQ stanine				0.090	41.84	0.028
Lowest	−0.706	−24.71	−0.205			
2	−0.580	−27.17	−0.177			
3	−0.477	−21.49	−0.151			
4	−0.369	−21.72	−0.121			
5	−0.263	−15.64	−0.089			
6	−0.149	−8.88	−0.052			
7	−0.075	−4.35	−0.027			
8	−0.023	−1.24	−0.008			
Education						
Basic	−0.006	−5.66	−0.002	−0.006	−5.72	−0.002
Vocational	−0.015	−14.27	−0.006	−0.015	−14.16	−0.005
Matricular	0.000	−0.04	0.000	0.000	−0.17	0.000

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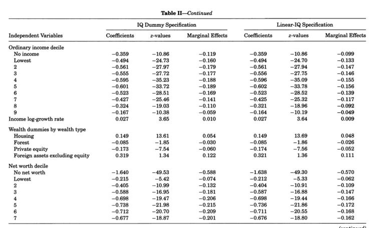


Table II—*Continued*

	IQ Dummy Specification			Linear-IQ Specification		
Independent Variables	Coefficients	z-values	Marginal Effects	Coefficients	z-values	Marginal Effects
Ordinary income decile						
No income	−0.359	−10.86	−0.119	−0.359	−10.86	−0.099
Lowest	−0.494	−24.73	−0.160	−0.494	−24.70	−0.133
2	−0.561	−27.97	−0.179	−0.561	−27.94	−0.147
3	−0.555	−27.72	−0.177	−0.556	−27.75	−0.146
4	−0.595	−35.23	−0.188	−0.596	−35.09	−0.155
5	−0.601	−33.72	−0.189	−0.602	−33.78	−0.156
6	−0.523	−28.51	−0.169	−0.523	−28.52	−0.139
7	−0.427	−25.46	−0.141	−0.425	−25.32	−0.117
8	−0.324	−19.03	−0.110	−0.321	−18.96	−0.092
9	−0.167	−10.38	−0.059	−0.164	−10.19	−0.049
Income log-growth rate	0.027	3.65	0.010	0.027	3.64	0.009
Wealth dummies by wealth type						
Housing	0.149	13.61	0.054	0.149	13.69	0.048
Forest	−0.085	−1.85	−0.030	−0.085	−1.86	−0.026
Private equity	−0.173	−7.54	−0.060	−0.174	−7.56	−0.052
Foreign assets excluding equity	0.319	1.34	0.122	0.321	1.36	0.111
Net worth decile						
No net worth	−1.640	−49.53	−0.588	−1.638	−49.30	−0.570
Lowest	−0.215	−5.42	−0.074	−0.212	−5.33	−0.062
2	−0.405	−10.99	−0.132	−0.404	−10.91	−0.109
3	−0.588	−16.95	−0.181	−0.587	−16.88	−0.147
4	−0.698	−19.47	−0.206	−0.698	−19.44	−0.166
5	−0.738	−21.98	−0.215	−0.736	−21.86	−0.172
6	−0.712	−20.70	−0.209	−0.711	−20.55	−0.168
7	−0.677	−18.87	−0.201	−0.676	−18.80	−0.162

(continued)

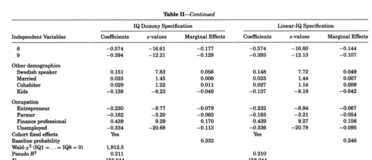


Table II—*Continued*

	IQ Dummy Specification			Linear-IQ Specification		
Independent Variables	Coefficients	z-values	Marginal Effects	Coefficients	z-values	Marginal Effects
8	−0.574	−16.61	−0.177	−0.574	−16.60	−0.144
9	−0.394	−12.21	−0.129	−0.393	−12.13	−0.107
Other demographics						
Swedish speaker	0.151	7.83	0.056	0.148	7.72	0.049
Married	0.023	1.45	0.008	0.023	1.44	0.007
Cohabiter	0.029	1.22	0.011	0.027	1.14	0.009
Kids	−0.138	−8.23	−0.049	−0.137	−8.18	−0.042
Occupation						
Entrepreneur	−0.230	−8.77	−0.079	−0.232	−8.84	−0.067
Farmer	−0.182	−3.20	−0.063	−0.183	−3.21	−0.054
Finance professional	0.439	9.29	0.170	0.439	9.27	0.156
Unemployed	−0.334	−20.68	−0.113	−0.336	−20.79	−0.095
Cohort fixed effects	Yes			Yes		
Baseline probability			0.332			0.246
Wald-χ^2 (IQ1 = … = IQ8 = 0)	1,912.5					
Pseudo R^2	0.211			0.210		
N	158,044			158,044		

Read. Table II is the paper's central result.

Key estimate	Value	Interpretation
Lowest vs highest IQ stanine	-20.5 pp	large participation gap
Linear IQ coefficient	0.090	monotone per-stanine effect
Math subscore coefficient	0.088	strongest subcomponent
Verbal subscore coefficient	0.070	significant
Logical subscore coefficient	0.065	significant

The text also emphasizes that the IQ effect is bigger than the income effect in the same regressions. The highest income subjects do participate more, but IQ has the more dramatic and more monotonic slope.

Economic message. The participation puzzle is not just about money. Cognitive skill is itself a major predictor of entry into risky asset markets.

5.4 Table III: affluent households

Read. Re-estimating the participation regressions in affluent subsamples gives:

Table III

IQ Scores and Stock Market Participation of Affluent Individuals

Table III reports summary data from probit regressions of stock market participation on IQ stanine dummies (or IQ score) and a host of control variables (described in the body of the paper) derived from the Finnish tax data and the Finnish census data set. The sample is restricted to the 10% of most affluent individuals in the data set. Panel A restricts the sample to the 10% of individuals with the largest ordinary income for 2000 as reported on their tax returns. Panel B restricts the sample to the 10% of individuals with the largest taxable net worth as reported on their year 2000 tax returns. Participation is a dummy variable that takes the value one for subjects who held mutual funds or individual stocks registered with the FCSD at the end of 2000. Pseudo R^2 and sample sizes are reported at the bottom of the table. Standard errors are clustered by zip code. For each of two specifications, the columns report coefficients from the probit regression, associated z-values, and marginal effects on participation probability (evaluated at the average value of the other regressors, except for IQ stanine dummies, which are evaluated at zero). The marginal effects for indicator variables indicate the shift in the participation probability when the indicator variable changes from zero to one. The dummy variable associated with the highest category—IQ stanine 9, university-level education, highest ordinary income, and taxable net worth in the highest decile—are omitted and serve as a benchmark. Taxable net worth deciles are computed after removing individuals with no taxable net worth; a dummy variable for no net worth identifies the latter individuals. The regressions also contain 30 (unreported) cohort fixed effects for birth years 1953 through 1982.

Panel A: Ordinary Income in Top 10% of the Distribution

	IQ Dummy Specification			Linear-IQ Specification		
Independent Variables	Coefficients	z-values	Marginal Effects	Coefficients	z-values	Marginal Effects
IQ stanine				0.059	8.00	0.023
Lowest	−0.396	−2.45	−0.157			
2	−0.708	−6.48	−0.275			
3	−0.372	−4.58	−0.147			
4	−0.273	−5.11	−0.108			

(continued)

Table III—*Continued*

Panel A: Ordinary Income in Top 10% of the Distribution

	IQ Dummy Specification			Linear-IQ Specification		
Independent Variables	Coefficients	z-values	Marginal Effects	Coefficients	z-values	Marginal Effects
5	−0.164	−4.11	−0.064			
6	−0.046	−1.16	−0.018			
7	−0.066	−1.71	−0.026			
8	−0.008	−0.19	−0.003			
Education						
Basic	−0.004	−1.37	−0.001	−0.004	−1.35	−0.002
Vocational	−0.015	−6.26	−0.006	−0.015	−6.22	−0.006
Matricular	−0.001	−0.25	0.000	−0.001	−0.19	0.000
Income log-growth rate	0.060	1.44	0.023	0.053	1.27	0.021
Wealth dummies by wealth type						
Housing	0.188	7.45	0.073	0.188	7.47	0.075
Forest	−0.024	−0.23	−0.009	−0.025	−0.24	−0.010
Private equity	−0.045	−0.82	−0.018	−0.049	−0.90	−0.020
Foreign assets excluding equity	0.901	1.70	0.280	0.915	1.75	0.299
Net worth decile						
No net worth	−1.311	−24.58	−0.467	−1.304	−24.43	−0.475
Lowest	−0.779	−8.84	−0.300	−0.772	−8.78	−0.293
2	−0.692	−7.45	−0.269	−0.696	−7.45	−0.267
3	−0.729	−8.21	−0.283	−0.724	−8.16	−0.277
4	−0.823	−9.03	−0.316	−0.812	−8.95	−0.307
5	−0.798	−11.67	−0.307	−0.791	−11.57	−0.300
6	−0.710	−9.15	−0.276	−0.704	−9.04	−0.271
7	−0.680	−8.69	−0.265	−0.672	−8.58	−0.260
8	−0.542	−7.97	−0.214	−0.537	−7.89	−0.211
9	−0.379	−5.95	−0.150	−0.374	−5.83	−0.148

(continued)

Table III—*Continued*

Panel A: Ordinary Income in Top 10% of the Distribution

	IQ Dummy Specification			Linear-IQ Specification		
Independent Variables	Coefficients	z-values	Marginal Effects	Coefficients	z-values	Marginal Effects
Other demographics						
Swedish speaker	0.138	3.00	0.053	0.131	2.84	0.051
Married	−0.010	−0.28	−0.004	−0.008	−0.23	−0.003
Cohabiter	0.080	1.39	0.031	0.074	1.28	0.029
Kids	−0.154	−4.27	−0.059	−0.152	−4.23	−0.060
Occupation						
Entrepreneur	−0.202	−3.79	−0.080	−0.221	−4.23	−0.088
Farmer	−0.092	−0.66	−0.036	−0.099	−0.71	−0.039
Finance professional	0.347	5.94	0.128	0.350	5.99	0.133
Unemployed	−0.116	−0.52	−0.046	−0.116	−0.52	−0.046
Cohort fixed effects	Yes			Yes		
Baseline probability			0.591			0.555
Wald-χ^2 (IQ1 = … = IQ8 = 0)	93.0					
Pseudo R^2	0.111			0.109		
N	15,413			15,413		

Panel B: Net Worth in Top 10% of the Distribution

	IQ Dummy Specification			Linear-IQ Specification		
Independent Variables	Coefficients	z-values	Marginal Effects	Coefficients	z-values	Marginal Effects
IQ stanine				0.094	6.70	0.023
Lowest	−0.826	−4.27	−0.225			
2	−0.628	−4.30	−0.157			
3	−0.496	−3.74	−0.117			
4	−0.595	−4.97	−0.147			

(continued)

Table III—*Continued*

Panel B: Net Worth in Top 10% of the Distribution

	IQ Dummy Specification			Linear-IQ Specification		
Independent Variables	Coefficients	z-values	Marginal Effects	Coefficients	z-values	Marginal Effects
5	−0.305	−2.96	−0.065			
6	−0.250	−2.47	−0.051			
7	−0.248	−2.33	−0.051			
8	−0.088	−0.67	−0.016			
Education						
Basic	−0.003	−0.55	0.000	−0.003	−0.60	−0.001
Vocational	−0.025	−5.81	−0.004	−0.025	−5.79	−0.006
Matricular	0.003	0.37	0.001	0.003	0.40	0.001
Ordinary income decile						
No income	−0.013	−0.08	−0.002	−0.014	−0.09	−0.003
Lowest	0.131	0.97	0.021	0.120	0.89	0.028
2	−0.189	−1.48	−0.037	−0.188	−1.47	−0.049
3	−0.246	−2.10	−0.050	−0.245	−2.10	−0.066
4	−0.221	−2.02	−0.044	−0.217	−1.97	−0.058
5	−0.261	−2.34	−0.053	−0.271	−2.44	−0.073
6	−0.291	−2.54	−0.060	−0.290	−2.54	−0.079
7	−0.241	−2.14	−0.048	−0.246	−2.21	−0.066
8	−0.132	−1.35	−0.025	−0.130	−1.35	−0.033
9	−0.091	−1.09	−0.017	−0.096	−1.15	−0.024
Income log-growth rate	0.009	0.23	0.002	0.010	0.25	0.002
Wealth dummies by wealth type						
Housing	−0.109	−1.04	−0.018	−0.110	−1.06	−0.026
Forest	0.130	1.03	0.022	0.131	1.04	0.030
Private equity	−0.070	−0.85	−0.013	−0.070	−0.85	−0.017

(continued)

Table III—*Continued*

Panel B: Net Worth in Top 10% of the Distribution

	IQ Dummy Specification			Linear-IQ Specification		
Independent Variables	Coefficients	z-values	Marginal Effects	Coefficients	z-values	Marginal Effects
Foreign assets excluding equity	−0.291	−0.79	−0.061	−0.274	−0.74	−0.075
Other demographics						
Swedish speaker	0.153	2.22	0.025	0.160	2.34	0.037
Married	0.080	0.87	0.014	0.080	0.88	0.019
Cohabiter	0.160	1.16	0.026	0.167	1.21	0.037
Kids	−0.286	−3.05	−0.051	−0.287	−3.08	−0.070
Occupation						
Entrepreneur	−0.137	−1.16	−0.026	−0.138	−1.18	−0.035
Farmer	−0.420	−3.13	−0.090	−0.419	−3.14	−0.117
Finance professional	0.781	2.95	0.084	0.778	2.93	0.124
Unemployed	0.091	0.52	0.015	0.087	0.50	0.020
Cohort fixed effects	Yes			Yes		
Baseline probability			0.899			0.842
Wald-χ^2 (IQ1 = … = IQ8 = 0)	49.6					
Pseudo R^2	0.144			0.142		
N	3,857			3,857		

Subsample	Lowest vs highest IQ gap
Top income decile	-15.7 pp
Top wealth decile	-22.5 pp

Economic message. IQ still matters where direct participation costs should matter least. This is strong evidence against the view that the main result is just low wealth in disguise.

5.5 Table IV: decomposition of the IQ gap

Read. The Fairlie-Blinder-Oaxaca decomposition finds:

Comparison	Total participation gap	Share explained by controls
Stanine 1 vs stanine 9	36.7 pp	63.5%
Stanine 2 vs stanine 8	29.4 pp	about 60%

For the 1-versus-9 comparison, the main channels are roughly:

- wealth: 8 pp,
- education: 6 pp,
- income: 6 pp,
- occupation and employment: 2 pp.

For the 2-versus-8 comparison, the pattern is similar: wealth, education, and income each explain about 5 to 6 percentage points of the gap.

Economic message. IQ matters both directly and indirectly. Much of the participation gradient is transmitted through wealth accumulation, education, and labor-market outcomes, but not all of it.

Table IV

Fairlie-Blinder-Oaxaca Decomposition of the Secondary Effects of IQ on Stock Market Participation

Table IV reports on a Fairlie-Blinder-Oaxaca decomposition. This analysis measures how much of the difference in high- and low-IQ individuals' stock market participation rates at the end of 2000 can be explained by differences in control variables such as education, income, and wealth. We first estimate a probit regression of a stock market participation dummy against all control variables, omitting the IQ regressor(s). We save the z-scores from this regression and translate them into predicted participation rates for different IQ groups. The decomposition technique computes the marginal effect of group mean differences for seven natural collections of the control variables. For a given stanine pairing, marginal effects are the sequence of changes in predicted participation rates obtained by sequentially changing each control variable's value from its group mean at the lower stanine to its mean at the higher stanine. Sequencing of the changes in the control variables are randomized, repeated, and averaged, and members are paired across the two stanines, to obtain marginal changes in participation rates and test statistics. Panel A reports on an analysis of participation rate differences between stanines 1 and 9. Panel B reports on stanines 2 and 8.

Panel A: Decomposition Estimates for IQ 1 versus IQ 9 Individuals		
	Decomposition Estimate, %	z-value
Education	6.03	48.1
Income	5.70	52.1
Asset class ownership	0.58	14.0
Wealth	8.29	131.2
Demographics	0.29	7.0
Profession and unemployment	1.83	28.5
Cohort	0.61	7.8
IQ=1 participation rate	9.80	
IQ=9 participation rate	46.52	
Explained difference in participation rates	23.33	
Unexplained difference in participation rates	13.39	

Panel B: Decomposition Estimates for IQ 2 versus IQ 8 Individuals		
	Decomposition Estimate, %	z-value
Education	4.57	47.6
Income	4.92	51.0
Asset class ownership	0.42	14.4
Wealth	5.99	135.4
Demographics	0.20	6.0
Profession and unemployment	1.32	29.6
Cohort	0.44	7.7
IQ=2 participation rate	13.07	
IQ=8 participation rate	42.52	
Explained difference in participation rates	17.88	
Unexplained difference in participation rates	11.57	

5.6 Table V: sibling control-function evidence

Read. In the brother sample, the control-function estimate gives:

Estimate	Value	Interpretation
Control-function IQ coefficient	0.253	strong positive own-IQ effect
Marginal effect	4.8 pp	economically meaningful
Sample	1,996 brother pairs	much smaller than baseline

Economic message. Correcting for endogeneity does not make the IQ effect disappear. If anything, the coefficient becomes larger, which is more consistent with attenuation from noisy IQ measurement than with upward omitted-variable bias.

5.7 Table VI: family random-effects probit

Read. Absorbing common family background still leaves a positive IQ effect:

Result	Estimate	Interpretation
Linear IQ coefficient	0.130 ($z = 4.54$)	within-family IQ still matters
Lowest vs highest dummy	-1.041 ($z = -2.62$)	strong low-IQ deficit
Sample	3,766 brothers	family effects absorbed

Economic message. The IQ gradient is not just a family fixed effect. Differences in IQ between brothers also predict differences in stock market participation.

Table V

Stock Market Participation Decisions Using a Control Function Approach to Estimation

Table V reports on a probit regression of stock market participation on a person's own IQ score, a host of control variables (described in the body of the paper) derived from the Finnish tax data and the Finnish census data set, and a residual from a first-stage OLS regression of one's own IQ score against his brother's IQ score and the control variables. The inclusion of the residual controls for an endogeneity problem that would arise if some unobservable controls were correlated with one's own IQ score. We identify 1,996 pairs of brothers using historical addresses and move-in and move-out dates for each subject in the Finnish tax data. Two males are identified as brothers if they lived together as children at the same address at the same time or moved at the same time. We also use transitivity to establish a sibling pair as described in the body of the paper. Participation is a dummy variable that takes on the value one for subjects who held mutual funds or individual stocks registered with the FCSD at the end of 2000. Pseudo R^2 and sample sizes are reported at the bottom of the table. Columns report coefficients from the probit regression, associated jackknife-estimated z-values, and marginal effects on participation probability (evaluated at the average value of the regressors). The dummy variable associated with the highest category—university-level education, highest ordinary income, and taxable net worth in the highest decile—are omitted and serve as a benchmark. Taxable net worth deciles are computed after removing individuals with no taxable net worth; a dummy variable for no net worth identifies the latter individuals. The regressions also contain 30 (unreported) cohort fixed effects for birth years 1953 through 1982.

Independent Variables	Coefficients	z-values	Marginal Effects
IQ stanine	0.253	5.21	0.048
Education			
Basic	0.000	0.00	0.001
Vocational	−0.019	−2.45	−0.004
Matricular	0.002	0.22	0.000
Ordinary income decile			
No income	0.092	0.49	0.036
Lowest	0.370	1.58	0.115
2	−0.247	−1.20	−0.021
3	−0.241	−1.34	−0.004
4	−0.238	−1.43	−0.005
5	−0.284	−2.08	−0.045
6	−0.106	−0.81	0.006
7	−0.233	−2.09	−0.028
8	−0.234	−2.17	−0.030
9	−0.222	−2.17	−0.024
Income log-growth rate	0.037	0.86	0.002
Wealth dummies by wealth type			
Housing	−0.282	−2.18	−0.017
Forest	−1.344	−2.37	−0.200
Private equity	−0.706	−2.66	−0.032
Foreign assets excluding equity			
Net worth decile			
No net worth	−2.238	−10.73	−0.352
Lowest	0.506	0.73	−0.213
2	0.891	3.45	−0.162
3	0.062	0.19	−0.178
4	0.288	0.81	−0.186
5	0.146	0.49	−0.185
6	−0.243	−0.90	−0.167
7	−0.197	−0.82	−0.129
8	−0.367	−1.49	−0.089
9	−0.550	−2.50	−0.079
Other demographics			
Swedish speaker	0.407	4.04	0.034
Married	−0.087	−0.32	−0.017
Cohabiter	0.280	0.44	0.015
Kids	−0.430	−0.77	−0.050
Occupation			
Entrepreneur	−0.553	−1.49	−0.023
Farmer	0.279	0.43	0.063
Finance professional	0.870	1.05	0.033
Unemployed	−0.412	−2.72	−0.082
1st-stage control variable	−0.176	−3.39	−0.033
Cohort fixed effects	Yes		
Baseline probability			0.110
Pseudo R^2	0.269		
N	3,992		

(continued)

Table V—*Continued*

Table VI

Random Effects Probit Analysis of Brothers' Stock Market Participation Decisions

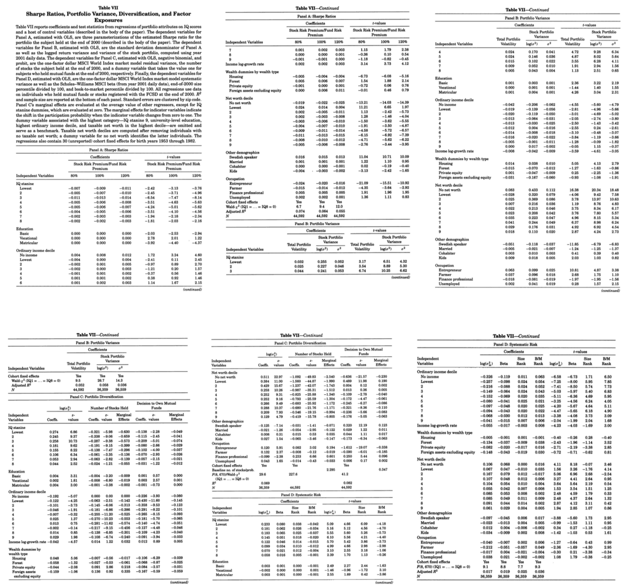
Table VI reports summary data from probit regressions of stock market participation on IQ scores and a host of control variables (described in the body of the paper) derived from the Finnish tax data and the Finnish census data set. Participation is a dummy variable that takes on the value one for subjects who held mutual funds or individual stocks registered with the FCSD at the end of 2000. The regressions are estimated using data on 1,996 brother pairs for whom we have both IQ scores and all control variables. Two males are identified as brothers if they lived together as children at the same address at the same time or moved at the same time. We also use transitivity to establish a sibling pair as described in the body of the paper. The regression controls for unobserved family background variables with family random effects. Pseudo R^2 and sample sizes are reported at the bottom of the table. For each of two specifications, the columns report coefficients from the probit regression and associated z-values. The dummy variable associated with the highest category—IQ stanine 9, university-level education, highest ordinary income, and taxable net worth in the highest decile—are omitted and serve as a benchmark. Taxable net worth deciles are computed after removing individuals with no taxable net worth; a dummy variable for no net worth identifies the latter individuals. The regressions also contain 30 (unreported) cohort fixed effects for birth years 1953 through 1982.

	IQ Dummy Specification		Linear-IQ Specification	
Independent Variables	Coefficients	z-values	Coefficients	z-values
IQ stanine			0.130	4.54
Lowest	−1.041	−2.62		
2	−0.599	−1.89		
3	−0.501	−1.77		
4	−0.530	−2.27		
5	−0.121	−0.56		
6	0.123	0.58		
7	−0.101	−0.44		
8	0.278	1.12		
Education				
Basic	−0.011	−0.80	−0.010	−0.73
Vocational	−0.047	−3.53	−0.045	−3.51
Matricular	−0.003	−0.25	−0.003	−0.24
Ordinary income decile				
No income	0.449	1.28	0.471	1.37
Lowest	0.669	1.69	0.645	1.65
2	−0.217	−0.61	−0.221	−0.63
3	−0.214	−0.72	−0.178	−0.61
4	−0.277	−1.05	−0.254	−0.98
5	−0.091	−0.40	−0.069	−0.31
6	0.048	0.22	0.038	0.18
7	−0.148	−0.78	−0.137	−0.74
8	−0.313	−1.78	−0.305	−1.75
9	−0.249	−1.51	−0.243	−1.50
Income log-growth rate	0.108	1.38	0.108	1.40
Wealth dummies by wealth type				
Housing	−1.866	−2.00	−1.885	−2.05

(continued)

Table VI—*Continued*

	IQ Dummy Specification		Linear-IQ Specification	
Independent Variables	Coefficients	z-values	Coefficients	z-values
Forest	−1.381	−2.99	−1.352	−2.98
Private equity				
Foreign assets excluding equity	−3.790	−10.62	−3.709	−10.67
Net worth decile				
No net worth	0.963	1.43	0.941	1.41
Lowest	1.854	1.94	1.877	1.98
2	0.245	0.42	0.288	0.50
3	0.992	1.63	0.975	1.65
4	0.558	1.03	0.613	1.15
5	−0.061	−0.13	−0.100	−0.22
6	−0.102	−0.25	−0.088	−0.22
7	−0.454	−1.09	−0.428	−1.04
8	−0.722	−1.88	−0.748	−1.98
9	0.642	3.33	0.630	3.32
Other demographics				
Swedish speaker				
Married	0.138	0.32	0.090	0.21
Cohabiter	0.977	1.04	0.821	0.89
Kids	−0.793	−0.99	−0.703	−0.90
Occupation				
Entrepreneur	−1.165	−2.31	−1.107	−2.24
Farmer	0.207	0.25	0.348	0.42
Finance professional	1.973	1.73	1.855	1.66
Unemployed	−0.748	−2.75	−0.746	−2.79
Cohort fixed effects	Yes		Yes	
Wald-χ^2 (IQ1 = ... = IQ8 = 0)	30.5			
N	3,766		3,766	



5.8 Table VII: Sharpe ratios, volatility, diversification, and factor exposures

Read. Table VII links IQ to portfolio quality among participants.

- **Panel A (Sharpe ratios).** When stock and fund risk premia are assumed equal, the Sharpe-ratio difference between the lowest and highest IQ-stanine participants is about **-0.009** (t = -3.13). For a portfolio with 30% annual volatility, the implied annual return disadvantage is roughly **21 to 33 basis points**, depending on the assumed stock premium.
- **Panel B (variance).** A one-stanine increase in IQ lowers logged stock-portfolio variance by about **3.5%**. High-IQ investors therefore achieve higher Sharpe ratios mainly through lower volatility.
- **Panel C (diversification).** High-IQ investors have lower residual variance, hold more individual stocks, and are more likely to hold mutual funds. Diversification improves systematically with IQ.
- **Panel D (systematic risk and factor exposures).** High-IQ investors have lower beta and lower systematic variance, while tilting toward smaller and higher book-to-market stocks.

Economic message. IQ is related not only to whether one enters the market, but also to whether one uses the market effectively.

6 Short summary

This paper studies whether cognitive ability helps explain stock market nonparticipation. Using Finnish administrative data that link military IQ tests to later stock and mutual-fund ownership, the authors show that participation rises strongly and monotonically with IQ. The effect is very large economically and survives controls for wealth, income, education proxies, demographics, occupations, and cohort effects.

The result is not confined to poorer households. Even among the top decile of income or wealth, low-IQ subjects are much less likely to participate. Sibling-based evidence further suggests that the result is not just a family-background story. A decomposition shows that wealth, education, and income are important channels through which IQ affects participation, but a large direct effect remains. Among participants, high-IQ investors build better portfolios: they diversify more, use mutual funds more, take less unrewarded risk, and earn higher Sharpe ratios.

7 Conclusion

Core conclusion. Cognitive ability is a first-order determinant of stock market participation and portfolio quality.

Economic intuition. Participation is not only about whether the expected equity premium is positive. It also depends on whether the household can understand the market well enough to construct a sensible risky portfolio. Low-IQ households appear to face a worse effective risk-return trade-off because they diversify less and make poorer portfolio choices.

Why the paper matters.

- It provides unusually clean evidence because it uses administrative ownership records instead of surveys.
- It shows that the participation puzzle survives among the affluent, where simple entry-cost stories are weakest.
- It warns empirical researchers that omitted cognitive ability can contaminate estimated effects of wealth, income, and education on participation.
- It suggests that models with universal participation miss an important source of heterogeneity.

Policy relevance. If low participation reflects low skill rather than irrational stubbornness alone, then forced market participation or tax incentives may not be enough. Default options, financial intermediation, and simple diversified products may matter more.

Bottom line. The paper moves the participation puzzle away from pure fixed-cost stories and toward a richer view in which cognitive skill affects both the decision to enter the stock market and the quality of the portfolio chosen after entry.